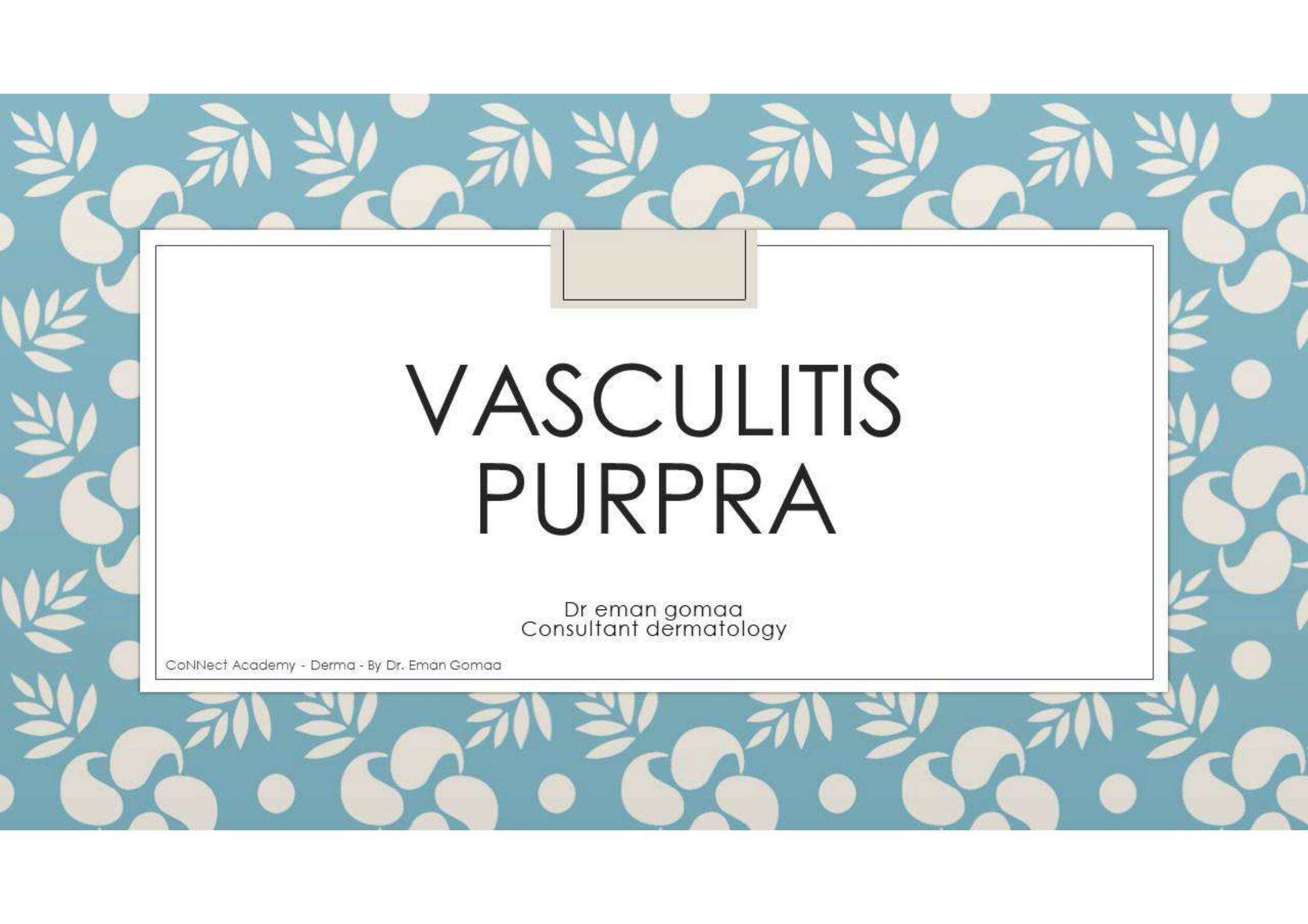
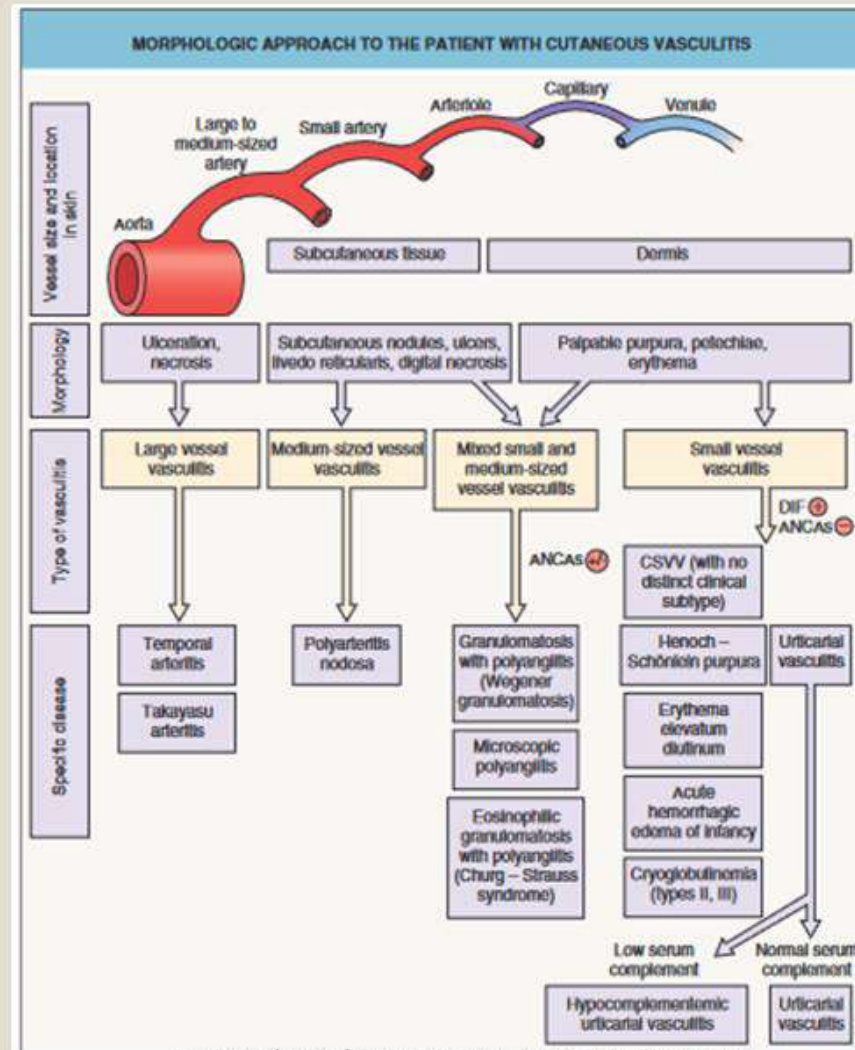




VASCULITIS PURPRA

Dr eman goma
Consultant dermatology

CoNNect Academy - Derma - By Dr. Eman Goma



Cryoglobulinaemic vasculitis

NB

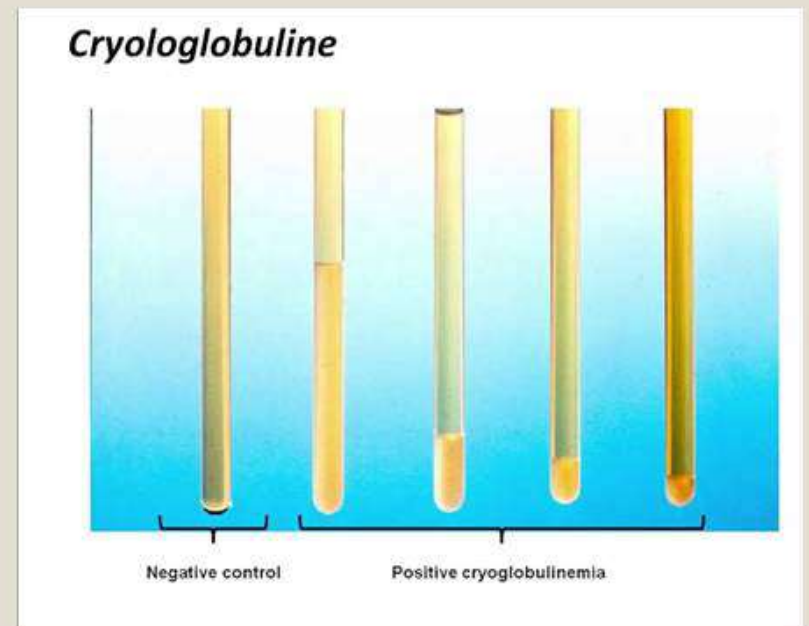
Cryoglobulins: abnormal immunoglobulins that precipitate spontaneously in vitro when serum is cooled to a temperature below 37° C

Cryoglobulinaemia: a condition characterized by the presence of circulating cryoglobulins.

- Cryoglobulins may be deposited as immune complexes in small vessels resulting in cryoglobulinaemic vasculitis



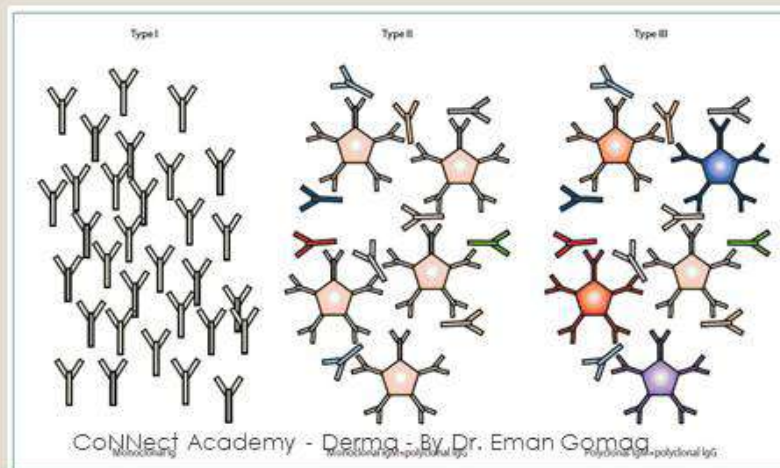
Courtesy: Headway - Dermatology - By Dr. Eman Gomaa



3 main types of cryoglobulinaemia

CLASSIFICATION OF CRYOGLOBULINS

Subtype	Molecular composition	Associations	Pathophysiology	Clinical manifestations
I	Monoclonal IgM > IgG*	Plasma cell dyscrasias, lymphoproliferative disorders	Vascular occlusion	Raynaud phenomenon, retiform purpura, gangrene, acrocyanosis
II**	Monoclonal IgM [†] (>IgG [†]) against polyclonal IgG	HCV >> HBV, HIV, autoimmune connective tissue diseases (e.g. rheumatoid arthritis), lymphoproliferative disorders (e.g. B-cell non-Hodgkin lymphoma, CLL)	Vasculitis	Palpable purpura, arthralgias, peripheral neuropathy, glomerulonephritis
III**	Polyclonal IgM [†] against polyclonal IgG			



- Cryoglobulinaemic vasculitis is Achronic systemic small-vessel vasculitis with multiple organ involvement (mainly the skin, joints, peripheral nerves and kidneys)
- **Mixed cryoglobulinaemia (Types II and III)**
- **hepatitis C virus (HCV)**, hepatitis B and HIV.
 - systemic LE,
- Immune complex mediated (type III hypersensitivity)

Cutaneous lesion :up to 90% of patients)

- Palpable purpura of the lower extremities, erythematous papules, ecchymoses and dermal nodules.
- Rarely, urticaria, livedo reticularis, necrosis, ulcerations and / or bullae.

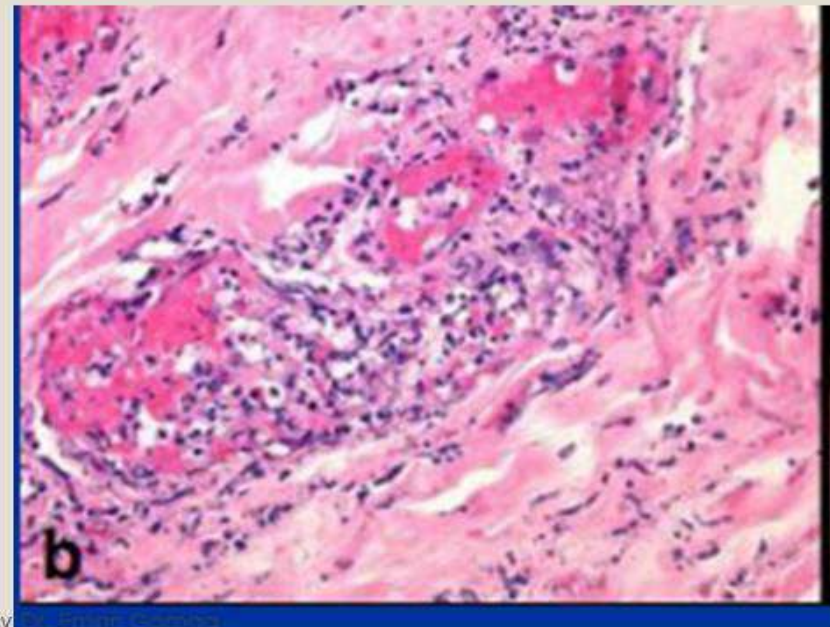
Systemic affection :**GAN**

Arthritis or arthralgias (70%)

- Peripheral (typically sensory) neuropathy (40%)
- Membrano-proliferative glomerulo-nephritis



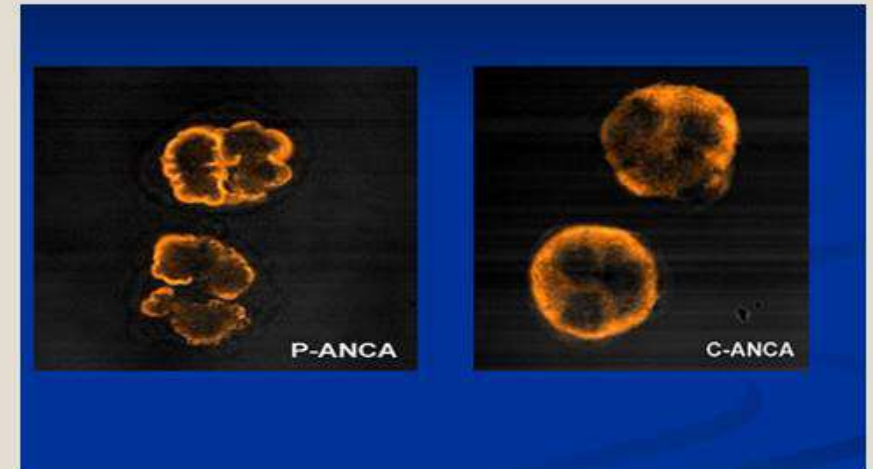
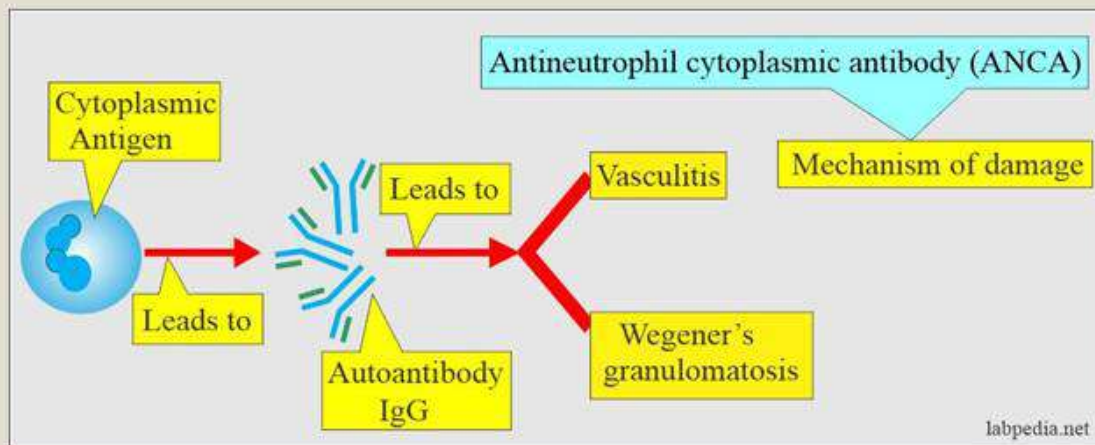
- HistopathoLOGY
 - Small vessel neutrophilic vasculitis
- DIF: granular deposits mainly of IgM and C3 in a vascular pattern in the papillary dermis.



TREATMENT

- Treatment of any underlying disease e,g
- HCV infections
Interferon-a plus antiviral (ribavirin)
- NON HCV :SRC
- combination of immunomodulatory agents (cyclophosphamide, rituximab) and glucocorticoid
- Plasma exchange

Small-vessel ANCA-associated vasculitis



ANTINEUTROPHIL CYTOPLASMIC ANTIBODY (ANCA)-ASSOCIATED VASCULITIDES

Disorder [No. of diagnostic criteria required]	Cutaneous and oral findings	Extracutaneous manifestations	Histologic findings [†]
Microscopic polyangiitis (MPA) [4]	• Palpable purpura • Erythematous macules, urticarial or purpuric plaques • Livedo racemosa, ulcers, splinter hemorrhages	• Renal: <i>glomerulonephritis</i> • Pulmonary: <i>capillaritis/hemorrhage, no asthma</i> • Neurologic: <i>mononeuritis multiplex</i> • Less often upper respiratory tract, cardiac, GI, and ocular involvement	• <i>Vasculitis of small (± medium-sized) vessels</i> • <i>No granulomas</i>
Granulomatosis with polyangiitis (GPA; formerly Wegener granulomatosis) [3; <i>anti-PR3/cANCA</i>]	• Palpable purpura • Friable, micropapular gingivae ('strawberry gums'), oral ulcers • PNGD* • Subcutaneous nodules • Pyoderma gangrenosum-like ulcers	• Upper respiratory: - <i>signs of inflammation, e.g. nasal ulcers, saddle nose, acute hearing loss, or chronic sinusitis/otitis/mastoiditis[†]</i> - <i>subglottic stenosis</i> • Pulmonary: <i>nodules/fixed infiltrates/cavities[†]</i> • Renal: <i>glomerulonephritis</i> • Ocular: <i>proptosis, scleritis</i> • Less often neurologic, GI, and cardiac involvement	• <i>Granulomatous inflammation</i> • <i>Vasculitis of small ± medium-sized vessels</i>
Eosinophilic granulomatosis with polyangiitis (EGPA; formerly Churg–Strauss syndrome [‡]) [4]	• Palpable purpura • PNGD* • Subcutaneous nodules • Urticarial plaques • Livedo racemosa, retiform purpura, ulcers	• Upper respiratory: <i>paranasal sinusitis[†], allergic rhinitis[§], nasal polyps[§]</i> • Pulmonary: - <i>asthma[§]</i> - <i>non-fixed infiltrate[†]</i> • Hematologic: <i>peripheral eosinophilia (>10%), elevated IgE</i> • Neurologic: <i>mono- or polyneuropathy</i> • Cardiac: <i>myo- or pericarditis</i> • Less often renal, GI, and ocular involvement	• <i>Extravascular eosinophils</i> • <i>Granulomatous inflammation</i> • <i>Vasculitis of small ± medium-sized vessels</i>

[†]May be observed in biopsies of the skin, mucosa, respiratory tract, kidney, or nerve.

*Palisaded neutrophilic and granulomatous dermatitis, which typically presents with umbilicated, crusted papulonodules on extensor surfaces (e.g. elbows) or the face (especially in GPA).

[†]Evidence via computed tomography (CT) represents a diagnostic criterion.

[‡]Occasionally triggered by leukotriene inhibitors and/or rapid discontinuation of corticosteroid therapy.

[§]Usually the initial manifestations.

RANGE OF FREQUENCIES OF ANCA IN ANCA-ASSOCIATED VASCULITIDES

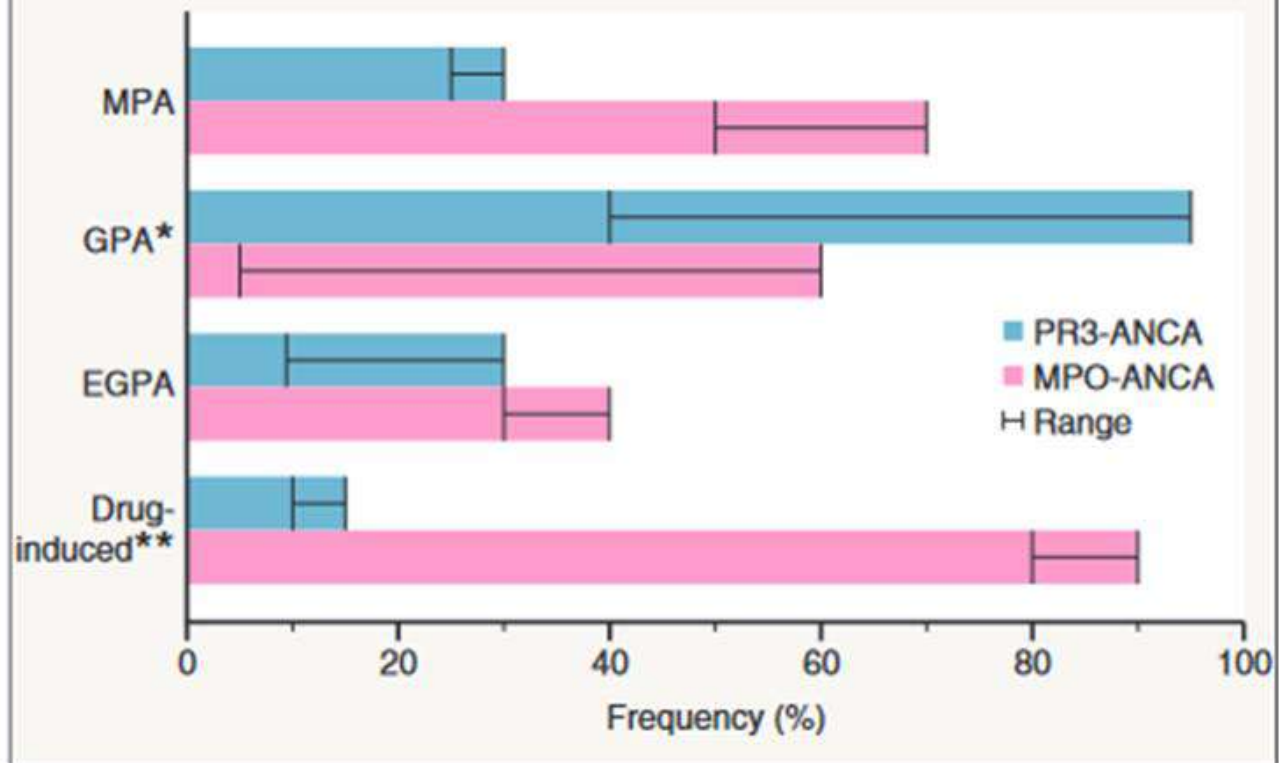




Fig. 24.15 Microscopic polyangiitis.
A Petechiae and purpuric macules on the lower extremity; some of the lesions have central hemorrhagic crusts. Histologically, leukocytoclastic vasculitis was seen.
B Petechiae and multiple purpuric macules with central necrosis on the plantar surface. *B, Courtesy, Caro Whitney Hannon, MD, and Robert Swenick, MD*

Microscopic Polyangiitis

*ANCA- Associated;
 Necrotizing pauci-immune, and no granulomas*

✓ Affects mostly small vessels, including venules and capillaries.

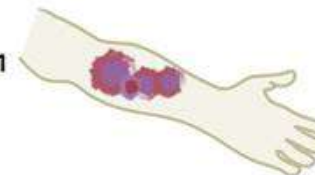
✓ Kidneys

- Glomerulonephritis with rapid progression to renal failure.



✓ Skin

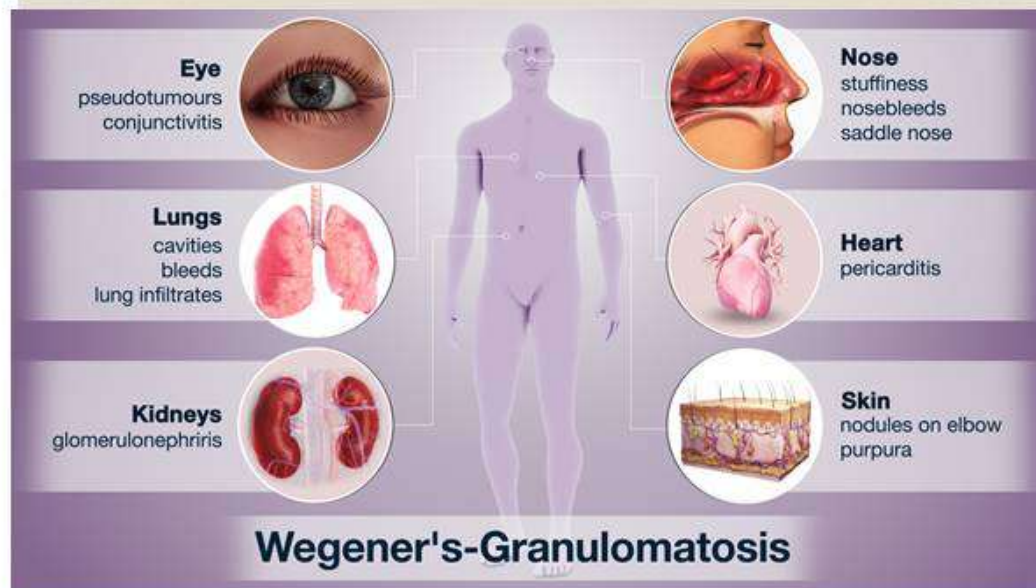
- Purpuric rash



✓ Lungs are less commonly involved

- Alveolar hemorrhage, fibrosis possible.

✓ Tends to occur in adults 50-60 y.o.



The standard treatment for patients with generalized GPA is systemic **corticosteroids** (e.g. 1 mg/kg/day of prednisone) in conjunction with oral daily **cyclophosphamide**



Fig. 24.17
Granulomatosis with polyangiitis (Wegener granulomatosis). **A** Ulceration on the leg, which may be misdiagnosed as pyoderma gangrenosum. **B** Ulceration of the tongue. **C** Subungual digital infarcts. **D** Palpable purpura of the legs due to small vessel vasculitis (leukocytoclastic vasculitis).

Churg-Strauss Syndrome



MP ACR criteria for the diagnosis of EGPA (Churg-Strauss syndrome)

1. Asthma (wheezing, expiratory rhonchi)
2. Eosinophilia >10% in peripheral blood
3. Sinusitis
4. Pulmonary infiltrates (may be transient)
5. Histological proof of vasculitis with extravascular eosinophils
6. Mononeuritis multiplex or polyneuropathy

The presence of four or more criteria is sufficient for diagnosis.

Over 80% of patients respond to treatment with corticosteroids alone



Fig. 24.18 Eosinophilic granulomatosis with polyangiitis (Churg–Strauss syndrome). **A** Palpable purpura of the buttocks due to small vessel vasculitis (leukocytoclastic vasculitis). **B** Purpuric dermal plaques of the palm that histologically demonstrated vasculitis of a small muscular artery. **C** Crusted, firm papules of the elbow. *C, Courtesy, Kalman Watsky, MD.*

POSSIBLE ADDITIONAL DIAGNOSTIC TESTING FOR PATIENTS WITH SUSPECTED ANCA-ASSOCIATED VASCULITIDES

	Chest X-ray and/or chest CT scan	Electromyogram/ nerve conduction studies	Lung, nerve or kidney biopsy	Upper respiratory tract or muscle biopsy	ECG, echocardiogram	CT scan of sinuses	Serum IgE level	Consider GI and ophthalmologic examination
Microscopic polyangiitis	√	√	√					
Granulomatosis with polyangiitis	√	√	√*	√*	√	√		√
Eosinophilic granulomatosis with polyangiitis	√	√	√		√	√	√	√

*Depends upon signs and symptoms.

Polyarteritis nodosa

- rare necrotizing vasculitis affecting mainly medium-sized arteries
- Types :
 - Classic PAN :multi-systemic affecting skin and other organ
 - Cutaneous PAN:single-organ vasculitis affecting the skin only

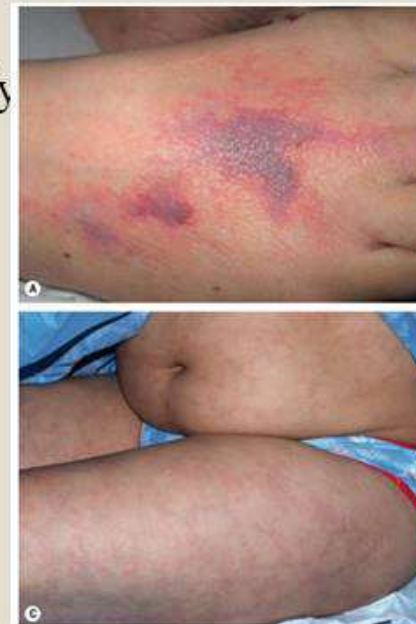


Fig. 24.19 Polyarteritis nodosa (PAN). A Retiform purpura of the dorsal foot in a patient with systemic PAN. B Livedo racemosa and subcutaneous nodules of the lower extremities in a patient with cutaneous PAN. C Livedo reticularis of the lower extremities and abdomen in an adolescent with cutaneous PAN.
C, Courtesy: Julie V. Schaller, MD

	CLASSIC	CUTANEOUS
	ADULT	children
	female :male 4:1	
	HBV	STREPT
C/P CUTANEOUS	Livedo reticularis , racemose , retiform purpra , painful sc nodules , Punched out ulcer , digital necrosis , Thigh , buttocks , hand , arm	
systemic	Fever , arthralgia , myalgia , abdominal pain , CHF , renal Aneurysm NO GN , NO LUNG	No
H p	Lcv at lower dermis , sc , early neutrophilic , late lymphocytic , histiocytic DIF :igM , c3 P ANCA 10- 20%	

TTT	Steroid 1 mg /kg + cyclophosphamide	NSAID Topical , intralesional steroid

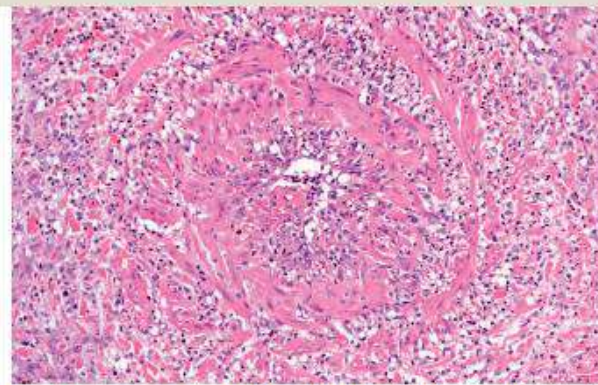


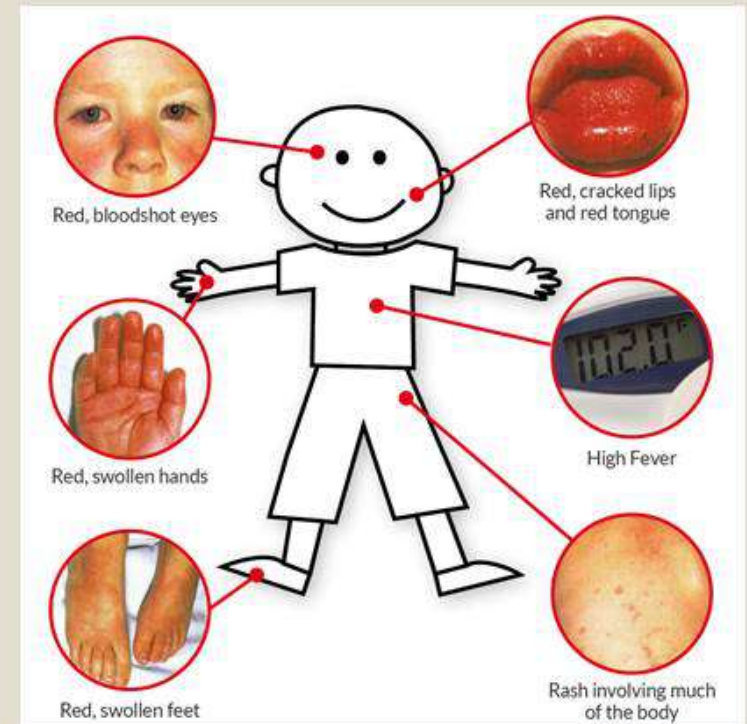
Fig. 24.20 Cutaneous polyarteritis nodosa – histologic features. Vasculitis of a medium-sized vessel situated deep in the dermis. Neutrophils, neutrophilic debris, and fibrin within the arteriolar wall are seen. Courtesy, Lorenzo Ceramchi MD.



Kawasaki disease

Mucocutaneous lymph node syndrome.

- KD is an acute febrile multi-system disease of unknown aetiology.
 - Chapel Hill Consensus nomenclature: medium-vessel vasculitis.
- in children < 2 years
- common in boys
- peak incidence in late winter to early spring.
- Japanese
- bacterial toxin that acts as a super-antigen.
Staphylococcus aureus and Streptococcus
- Genetic susceptibility and immune dysregulation
- reduces the efficiency of ITPKC → immune hyperreactivity. T cell



Clinical features

Kawasaki Disease Diagnosis Criteria



knowmedge

Mnemonic: "CREAM"

Criteria

Fever for greater than 5 days



4 out of 5 of the following

C	Conjunctivitis (non-exudative)
R	Rash (polymorphous non-vesicular)
E	Edema (or erythema of hands or feet)
A	<u>Adenopathy</u> (cervical, often unilateral)
M	Mucosal involvement (erythema or fissures or crusting)

Intellectual Property of Knowmedge.com

Signs & Symptoms of Kawasaki Disease



Images courtesy of the Kawasaki Foundation



- Systemic affection:
 - Cardiac involvement:
coronary artery aneurysms (ectasias) : 10-25%
- **OTHERS**
- **INVESTIGATION :**
 - Laboratory findings:
 - **CBC:** Leukocytosis (with neutrophil predominance), anemia and (less often) thrombocytopenia. By the 2nd - 3rd week, thrombocytosis is often present (i.e. late post-acute phase).
- **ECG , ECO**

- PROGNOSIS:

- Fever resolve after 1-2 weeks
 - 25% of cases with myocarditis can be life threatening

◦ TREATMENT:

1. **Intravenous immunoglobulin (IVIg);**

- The first-line therapy.
- A single infusion of 2 g/kg over 10-12 hours

2. **Aspirin:**

- 80-100 mg/kg daily divided into four doses
for 14 days>>>>then 3-5 mg /kg maintenance dose
for 8 weeks

Large size vasculitis

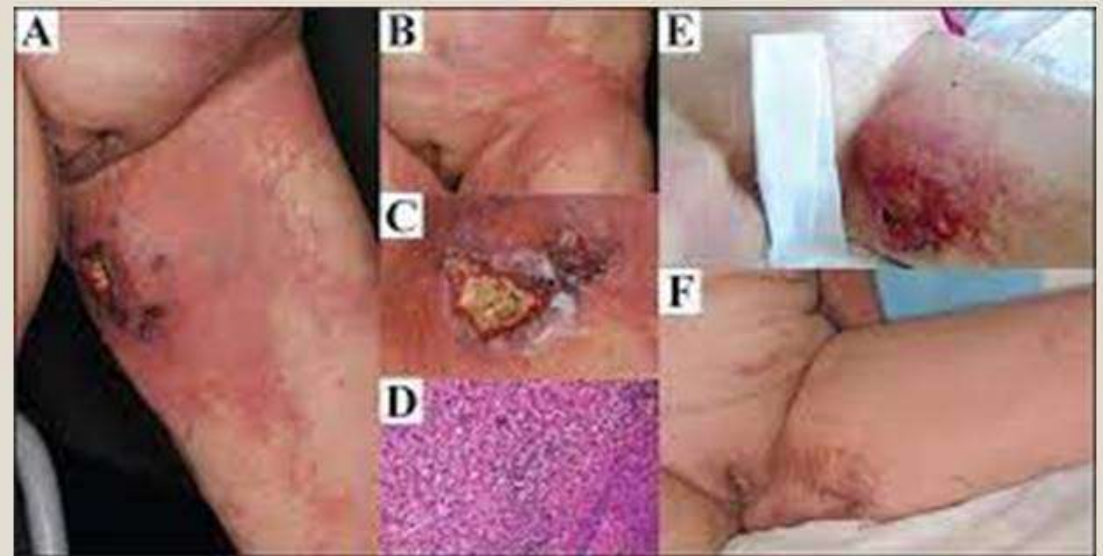
Temporal arteritis
granulomatous arteritis affecting the
aorta and/or its major branches

fever and weight loss
Headach , facial pain
Visual loss , deafness
Skin infarction

Takayasu arteritis
granulomatous arteritis affecting the
aorta and its major branches
EN , PG , erythema induratum
Hypertension, coronary ischemia



Fig. 24.21 Temporal arteritis. Two ulcerations of the scalp in association with alopecia and scarring in an elderly patient. *Courtesy, Steve Feldman, MD.*



behcet disease

Adamantiades-Behcet disease

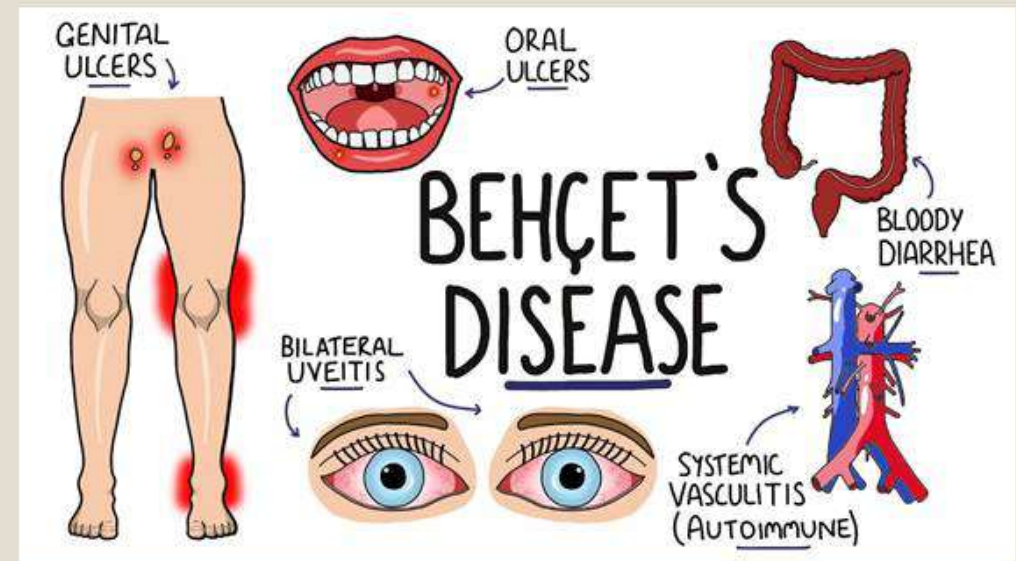
multi-systemic inflammatory disease of unknown aetiology

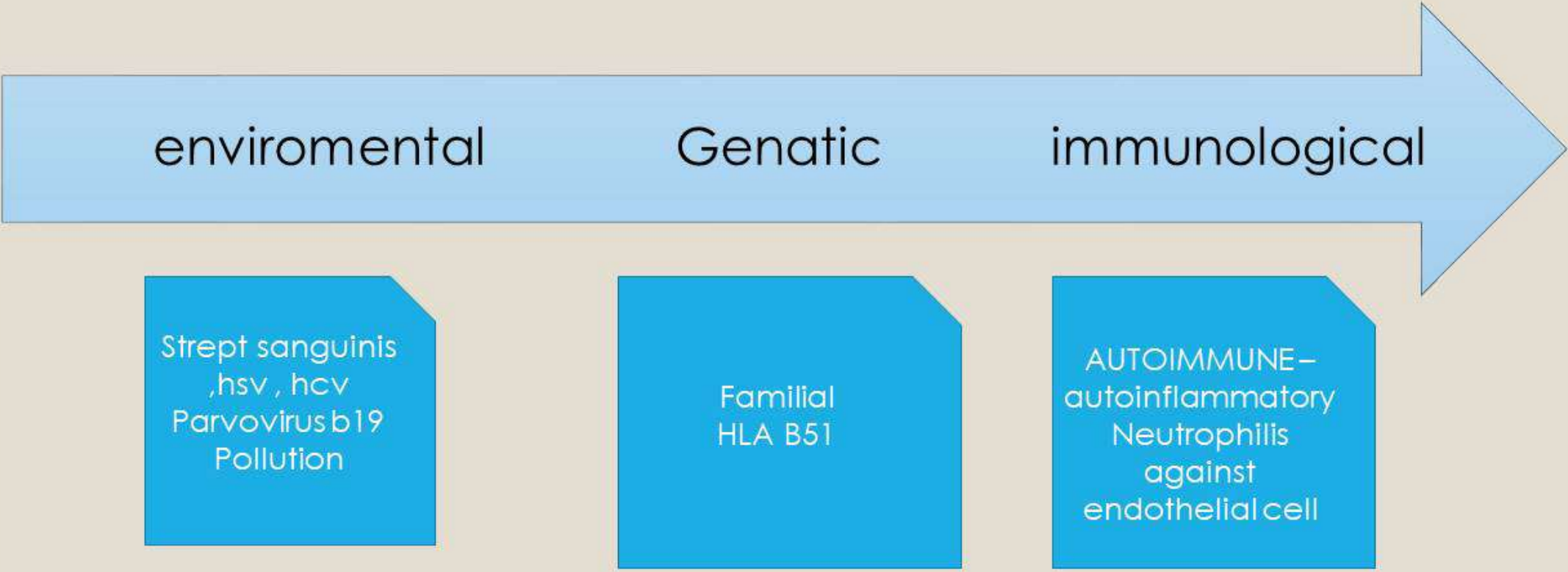
- . neutrophilic dermatosis
- cutaneous vasculitis associated with systemic disease or variable vessel size

in 2nd - 3rd decades.

A male predominance in Arab populations.

- A female predominance in Japan,





enviromental

Strept sanguinis
,hsv , hcv
Parvovirus b19
Pollution

Genatic

Familial
HLA B51

immunological

AUTOIMMUNE-
autoinflammatory
Neutrophilis
against
endothelial cell



– mucocutaneous lesions. **A, B** Oral aphthosis involving the tongue and lip; the ulcerations can be deep. These ulcers may be diagnosed as infectious (e.g. due to herpes simplex virus) or neoplastic rather than inflammatory. **C** Aphthae of the vulva and inguinal region. **D** Pathergy – a papulopustule appeared at the site of insertion of an intravenous cathether.



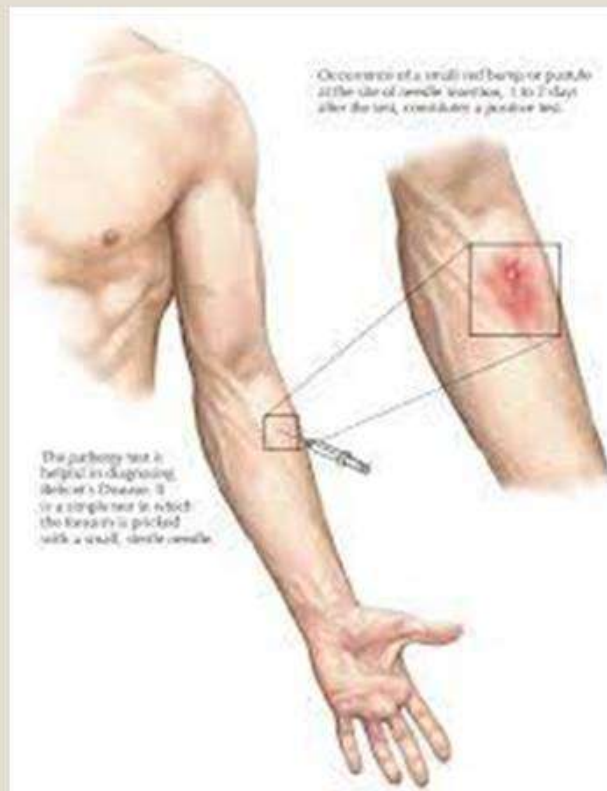
INTERNATIONAL STUDY GROUP CRITERIA FOR THE DIAGNOSIS OF BEHÇET DISEASE	
Major criterion	Required features
Recurrent oral ulceration	Aphthous (idiopathic) oral ulceration observed by physician or patient, recurring at least three times in a 12-month period
PLUS ANY TWO OF THE FOLLOWING MINOR CRITERIA:	
Recurrent genital ulceration	Aphthous genital ulceration or scarring, observed by physician or patient
Eye lesions	Anterior or posterior uveitis; cells in the vitreous by slit lamp examination; or retinal vasculitis observed by ophthalmologist
Cutaneous lesions	Erythema nodosum-like lesions observed by physician or patient; papulopustular lesions or pseudofolliculitis; or characteristic acneiform nodules observed by physician in postadolescent patient not on corticosteroids
Pathergy test*	Interpreted at 24–48 h by physician
*Pathergy test is performed on the flexor forearm by obliquely inserting a 20–22-gauge sterile hypodermic needle to a depth of 5 mm ± an intradermal injection of 0.1 ml of normal saline. A positive reaction is defined as the development of a papule or pustule.	

Behcet's Disease (2013) provide the most accurate diagnosis.

1. Oral aphthosis (recurrent, 2 points)
2. Genital aphthosis (recurrent, 2 points)
3. Ocular lesions (recurrent, 2 points)
4. Skin lesions (recurrent, 1 point)
5. Central nervous system lesions (1 point)
6. Vascular manifestations (1 point)
7. Positive pathergy test (1 point)

• A score > 4 indicates Adamantiades-Behcet disease

Pathergy test



Inserting a sterile needle to a depth of 5 mm with / without an intra-dermal injection of 0.1 ml of saline

. The test is interpreted after 24-48 hours.

- Positive test: development of an erythematous papule (>2 mm) or pustule at the site of skin prick.

POSITIVE IN :

pyoderma gangrenosum

, Sweet syndrome,

rheumatoid arthritis,

Crohn disease,

genital herpes infection,

, pyodermatitis- pyostomatitis vegetans

PAPA syndrom

JANT

SYSTEMIC MANIFESTATIONS OF BEHÇET DISEASE	
Ocular (leading cause of morbidity)^{31,33,34} • Occurs in 90% of patients; favors men, in whom it is more severe • Can be painful and may lead to blindness • Retinal vasculitis (more frequently associated with blindness) • Posterior uveitis (most characteristic ocular finding) • Anterior uveitis (Fig. 26.16), hypopyon • Secondary glaucoma, cataracts • Conjunctivitis, scleritis, keratitis, vitreous hemorrhage, optic neuritis	Neurologic • Usually appears later during the evolution of the disease • Associated with a poor prognosis • Acute meningo-encephalitis which may resolve spontaneously • Cranial nerve palsies • Brainstem lesions that can induce swallowing difficulties, laughter and crying • Pyramidal or extrapyramidal signs
Joints • Approximately 50% of patients develop arthritis • In majority (~80% of patients), duration of attacks is <2 months • Mono- or polyarthritic and non-erosive • Most commonly knees, wrists and ankles	Vascular • Aneurysmal or occlusive arterial disease • Superficial or deep venous thrombosis
Gastrointestinal • Abdominal pain and/or hemorrhage may be difficult to distinguish from IBD (see Table 26.13) • Ulcerations* develop within the small bowel (in particular the ileocecal region) as well as the transverse and ascending colon and esophagus; perforation can occur	Cardiopulmonary • Coronary arteritis, valvular disease, myocarditis • Recurrent ventricular arrhythmias • Pulmonary artery aneurysms
	Renal • Glomerulonephritis

*Resemble anogenital aphthae.

DD BS PE

- **Oculo-muco-cutaneous syndromes)'.
 - Behcet disease
 - Sweet syndrome
 - Erythema multiforme
 - Pemphigus
 - Cicatricial pemphigoid
 - Reactive arthritis**

TREATMENT

THERAPEUTIC LADDER FOR THE TREATMENT OF BEHÇET DISEASE 		
Treatment	Dose	Level of evidence
<i>Mucocutaneous disease</i>		
Viscous lidocaine, topical sucralfate and other symptomatic treatments		3
Topical and inhalant corticosteroids		2
Intralesional corticosteroids		3
Colchicine	0.6 mg po thrice daily	1
Dapsone	50–150 mg po daily (or sulfapyridine equivalent)	1
Combination oral colchicine plus dapsone		3
Apremilast	30 mg po twice daily	1
<i>Severe mucocutaneous disease</i>		
Thalidomide	50–150 mg po nightly	1
Methotrexate	2.5–25 mg po or IM weekly	3
Prednisone, intermittent with taper	40–80 mg po daily usual starting dose	2
Interferon- α -2a	Variable dose (3–9x 10 ⁶ IU sc 3x weekly)	1
TNF- α inhibitors	Infliximab 5 mg/kg IV at weeks 0, 2 and 6; adalimumab 80 mg sc as initial dose then 40 mg sc weekly or every other week; etanercept 50–100 mg sc weekly (in 1 or 2 doses)	1 (etanercept)
<i>Systemic disease</i>		
Prednisone	60–120 mg po daily usual starting dose (including split-dose), with taper to alternate days	2
Methylprednisolone acetate	40 mg IM 3x weekly	2
Azathioprine	50–100 mg po twice daily	1
Cyclophosphamide	Variable oral dose (50–200 mg daily) or IV pulse (500–1000 mg monthly then 500–1000 mg every 3 months)	3
Chlorambucil	4–6 mg po daily	1
Mycophenolate mofetil	1–1.5 g po twice daily	3
Cyclosporine	Variable doses (2–10 mg/kg po daily)	1

purpura

- purpura is the discoloration of (visible hemorrhage into) the skin or mucous membranes that does not blanch on pressure
- Petechiae: small (< 4 mm)
 - Purpura: medium (5 - 9 mm)
 - Ecchymoses (bruises): large (> 1 cm)



purpura

Disorders of the platelets:

- thrombocytopenia
- Abnormal platelet function
- Thrombocytosis

Disorders of the (coagulation) factors

- Factor deficiency / dysfunction:
 - (e.g. hemophilia) o
 - Drugs e.g. anticoagulants
 - Metabolic e.g. vitamin K deficiency,
 - DIC and purpura fulminans

Disorders of the blood vessels:

- Inflammatory purpura:
 - Vasculitis
 - Capillaritis
- Non-inflammatory purpura:
 - Intravascular
 - : Vessel wall disorders
 - External causes
 - OTHERS

Pigmented purpuric dermatosis capillaritis

- Pigmented purpuric dermatoses (PPD) are a group of chronic disorders with the following characteristics:
 - petechial hemorrhage (purpura) with yellow-brown background due to marked hemosiderin deposition usually affecting lower limbs.
 - asymptomatic without any systemic affection
- chronic inflammation of superficial papillary dermal vessels, usually capillaries (hence the term capillaritis) —•hemorrhage
- Histopathology: ALL these diseases show
- endothelial cell swelling, perivascular lymphocytic infiltrate,
- extravasation of RBCs, hemosiderin deposits,

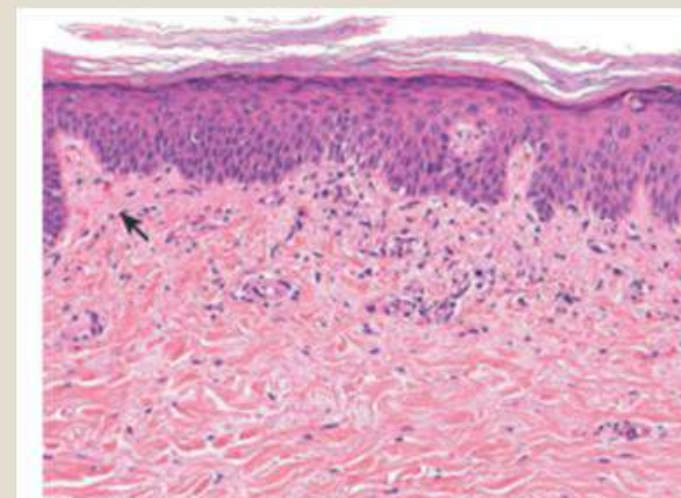


Fig. 22.8 Pigmented purpuric eruption – histopathologic features. Minimal perivascular lymphocytic infiltrate with extravasation of red blood cells (arrow) and hemosiderin deposits. Courtesy: Lorenzo Cesoni, MD.

TYPES OF PIGMENTED PURPURIC DERMATOSES			
Type	Age/Sex	Morphology	Location
Schamberg disease	• Most common form • Can occur in children • Peak frequency in middle-aged to older men	• Yellow-brown patches with an oval to irregular outline (Fig. 22.6A) • Pinpoint petechiae within patches – likened to “cayenne pepper” • Successive crops	• Favors lower legs • Can also involve thighs, buttocks, trunk and arms
Purpura annularis telangiectodes of Majocchi	• Uncommon • Adolescents and young adults, especially women	• 1–3 cm annular plaques that may slowly expand • Punctate telangiectasias and petechiae within the border (Fig. 22.7A)	Trunk, proximal lower extremities
Pigmented purpuric lichenoid dermatitis of Gougerot and Blum	• Rare • Middle-aged to older men	• Admixture of two types of lesions: (1) Schamberg-like (2) Purpuric red-brown lichenoid papules • Chronic • Occasionally pruritic	On lower extremity, may be confused with cutaneous small vessel vasculitis
Eczematid-like purpura of Doucas and Kapetanakis	• Rare • Middle-aged to older men	• Scaly petechial or purpuric macules, papules and patches (Fig. 22.7B) • Pruritic	Primarily lower extremities
Lichen aureus	• Rare	• Solitary patch • Color varies from golden to rust to purple-brown	Usually lower extremity, often over a perforator vein
Linear pigmented purpura	• Rare • Can occur in children and adolescents	• Linear array, usually unilateral (Fig. 22.7C) • Individual lesions similar in appearance to lichen aureus or Schamberg disease	Often a single extremity
Granulomatous pigmented purpura	• Very rare	• Brown patches with superimposed hemorrhagic papules*	Lower legs

*Histologically, hemorrhagic granulomatous inflammation in upper dermis.

schamberg's disease



schamberg's disease



purpura annularis telangiectodes of majocchi



purpura annularis telangiectodes of majocchi



LICHEN AUREUS





Fig. 22.7 Less common forms of pigmented purpuric eruptions (capillaritis). **A** Purpura annularis telangiectodes of Majocchi characterized by annular plaques with cayenne pepper petechiae in the border. **B** Eczematoid-like purpura that often presents with pruritus. **C** Linear pigmented purpuric dermatosis in which there is a linear array of yellow-brown macules and patches with superimposed petechiae and small, red-brown purpuric papules on the arm of a child. The eruption had been present for several months and was limited to this segment of skin. *B Courtesy Robinson-Worke, MD; C Courtesy John V.*

- treatment:
 - Explanation and reassurance without
 - Supportive measures e.g. elastic stockings.
 - A scorbic acid (500 mg twice daily)

Other causes of purpura

- **1- malignant atrophic papulosis (DEGOS disease)**
 - Skin
 - , GIT
 - , CNS

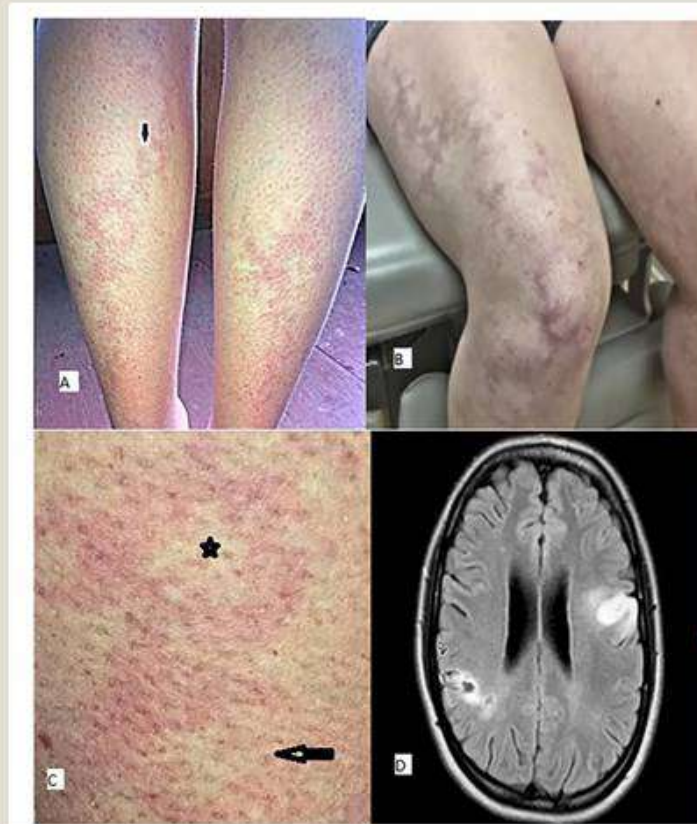


- **ATROPHIE BLANCHE (LIVEDOID VASCULOPATHY)**
 - **Ulcer +**
 - **Livedo reticularis**
 - **Retiform purpura**



- **Sneddon syndrome**

- Skin
- Brain
- Livedo reticularis
- Hypertension
- strok



- **Calcific uremic arteriolopathy**
 - Metastatic classification
 - Painful purpuric plaque with
 - Retiform stellate pattern
 - Abdomen , thigh , hip



**THANK
YOU**